定理（任意一族环的直积的泛性质刻画）$(R_i)_{i\in I}$ 是任意的一族环，

R 是一个环，$(p_i: R \longrightarrow R_i)_{i\in I}$ 是一族映射，满足：

★ 每个 p_i 都是环同态

★ 对 $\forall i\in I$，任取 $r_i\in R_i$，存在 $\alpha\in R$，使得对 $\forall i\in I$，都有 $p_i(\alpha)=r_i$

★ 对 $\forall x, y\in R$，若对 $\forall i\in I$，有：$p_i(x)=p_i(y)$，则有：$x=y$

R' 是另一个环，$(p_i': R' \longrightarrow R_i)_{i\in I}$ 是另一族映射，满足：

★ 每个 p_i' 都是环同态

★ 对 $\forall i\in I$，任取 $r_i\in R_i$，存在 $\alpha'\in R'$，使得对 $\forall i\in I$，都有 $p_i'(\alpha')=r_i$

★ 对 $\forall x, y\in R'$，若对 $\forall i\in I$，有：$p_i'(x)=p_i'(y)$，则有 $x=y$

则有：存在唯一的映射 $\varphi: R \longrightarrow R'$，使得对 $\forall i\in I$，下图交换：

$$\begin{tikzcd}
R \arrow[rr, "\varphi"] \arrow[dr, "p_i"'] & & R' \arrow[dl, "p_i'"] \\
& R_i &
\end{tikzcd}$$

且映射 φ 是环同构.

Proof: 首先，忘记所有环的环结构，仅仅把所有的环看作普通的集合，

可以得到：存在唯一的映射 $\varphi: R \longrightarrow R'$，使得对 $\forall i \in I$，下图交换：

$$\begin{tikzcd}
R \arrow[rr, "\varphi"] \arrow[dr, "p_i"'] & & R' \arrow[dl, "p_i'"] \\
& R_i &
\end{tikzcd}$$

且映射 φ 是双射.

下证 $\varphi: R \longrightarrow R'$ 是环同态.

对 $\forall x, y \in R$，有：

$\because x \in R$，$(p_i: R \longrightarrow R_i)_{i \in I}$ 是一族映射

$\therefore$ 对 $\forall i \in I$，$p_i(x) \in R_i$ $\quad \therefore \exists \alpha' \in R'$，s.t. 对 $\forall i \in I$，都有 $p_i'(\alpha') = p_i(x)$

$\therefore \varphi(x) = \alpha' \in R'$

$\because y \in R$，$(p_i: R \longrightarrow R_i)_{i \in I}$ 是一族映射

$\therefore$ 对 $\forall i \in I$，$p_i(y) \in R_i$ $\quad \therefore \exists \beta' \in R'$，s.t. 对 $\forall i \in I$，都有 $p_i'(\beta') = p_i(y)$

$\therefore \varphi(y) = \beta' \in R'$

$\because R$ 是环 $\quad \therefore x + y \in R \quad \therefore \varphi(x+y) \in R' \quad \because xy \in R \quad \therefore \varphi(xy) \in R'$

$\because R'$ 是环 $\quad \therefore \varphi(x) + \varphi(y) \in R'$，$\varphi(x)\varphi(y) \in R'$

$\varphi(x) + \varphi(y) \in R'$，$\varphi(x+y) \in R'$，对 $\forall i \in I$，有：

$p_i'(\varphi(x) + \varphi(y)) = p_i'(\alpha' + \beta') = p_i'(\alpha') + p_i'(\beta') = p_i(x) + p_i(y)$

$= p_i(x+y) = (p_i' \circ \varphi)(x+y) = p_i'(\varphi(x+y)) \quad \therefore \varphi(x) + \varphi(y) = \varphi(x+y)$

$\therefore \varphi(x+y) = \varphi(x) + \varphi(y)$

$\varphi(xy) \in R'$，$\varphi(x)\varphi(y) \in R'$，对 $\forall i \in I$，有：

$p_i'(\varphi(xy)) = (p_i' \circ \varphi)(xy) = p_i(xy) = p_i(x)p_i(y) = p_i'(\alpha')p_i'(\beta')$

$= p_i'(\alpha'\beta') = p_i'(\varphi(x)\varphi(y))$

$\therefore \varphi(xy) = \varphi(x)\varphi(y)$

$\because 1_R \in R$，$(p_i : R \longrightarrow R_i)_{i\in I}$ 是一族映射

$\therefore$ 对 $\forall i \in I$，$p_i(1_R) \in R_i$ $\quad \therefore \exists \gamma' \in R'$，s.t. 对 $\forall i \in I$，都有 $p_i'(\gamma') = p_i(1_R)$

$\therefore \varphi(1_R) = \gamma' \in R'$

$\varphi(1_R) \in R'$，$\quad 1_{R'} \in R'$，对 $\forall i \in I$，有：

$p_i'(\varphi(1_R)) = (p_i' \circ \varphi)(1_R) = p_i(1_R) = 1_{R_i} = p_i'(1_{R'})$

$\therefore \varphi(1_R) = 1_{R'}$

$\therefore \varphi : R \longrightarrow R'$ 是环同态.

$\therefore \varphi : R \longrightarrow R'$ 是环同构. $\quad \square$